The Micro-Task Market for Lemons: Data Quality on Amazon's Mechanical Turk

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Abstract

Response-quality screens flag a substantial share of records in three surveys collected on Amazon's Mechanical Turk. IP flags decrease from 20.3% in August 2018 to 9.8% in July 2020, while endorsement of multiple uncommon traits rises from 6.2% to 21.0%. Combining the screens flags 24.6%, 37.9%, and 27.6% in the August 2018, June 2020, and July 2020 surveys. In a randomized economic-evaluation experiment, an out-party rather than an in-party cue lowers unemployment evaluations by -0.096 on a zero-to-one scale in the full sample and by -0.125 among unflagged records. The full-sample contrast is 23.0% smaller (95% paired IP-cluster bootstrap interval: 4.2% to 38.7%). Screening changes the estimated contrasts, but does not identify the causal effect of dishonesty: IP metadata, reported uncommon traits, and date conventions are imperfect measures of response quality. These findings support reporting how recruitment and screening choices affect sample composition and inference, with the evidence confined to the archived surveys.

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1 The question and the evidence

Mechanical Turk lowers the cost of recruiting survey participants, but researchers have limited information about the quality of the responses they buy. Across three surveys, IP flags fall from 20.3% to 9.8%, while the share endorsing multiple uncommon traits rises from 6.2% to 21.0%. In the June experiment, the full-sample partisan-cue effects are smaller than those among unflagged respondents. Response screening therefore matters both for describing the sample and for estimating experimental effects.

Online recruitment broadened participation beyond university subject pools [Sears, 1986, Berinsky et al., 2012, Paolacci et al., 2010]. We examine the quality of those responses using IP metadata, response times, uncommon-trait questions, self-reported sincerity, and randomized experiments. The studies were fielded in August 2018, June 2020, and July 2020.

These indicators capture different aspects of response quality. Shared networks, misunderstanding, and valid uncommon experiences can produce flags alongside inattention or misrepresentation. We report the indicators separately and together, then compare experimental estimates across the resulting groups.

2 Incentives and what the design identifies

A micro-task market connects requesters who pay for tasks with workers who complete them. Ratings can encourage effort when requesters can verify quality. Survey opinions and personal experiences are harder to verify than a classification task with a known answer. A requester may approve questionable work because checking it costs time, because the response could be sincere, or because an incorrect rejection would treat the worker unfairly. The resulting approval rate is an imperfect measure of survey response quality [Akerlof, 1970, Peer et al., 2014].

A managed panel can accumulate information across repeated interactions with its participants. That difference motivates comparison with MTurk, but it is not experimentally manipulated here. The studies differ in recruitment, calendar time, questionnaire, pay, and eligibility restrictions. Their comparisons describe the samples obtained under those conditions. They do not isolate the causal effect of a platform, an approval threshold, or changes in Amazon's policies. Similarly, reported task experience is associated with measured response patterns; it was not randomly assigned.

3 Surveys, measures, and denominators

Study 1 was fielded on August 17, 2018, advertised as a ten-minute survey, and paid \$0.60 per completed task. Study 2 was fielded in June 2020, advertised as a fifteen-minute survey, and paid \$0.75. Study 3 was fielded in July 2020, advertised as a five-minute survey, and paid \$0.35. Studies 1 and 3 required a task approval rate of at least 95%; Study 2 did not. Each sought participants located in the United States. Table 1 reports the archived record counts; availability varies across survey items and models. Study 1 records are paid assignments; survey analyses use the available answers attached to those assignments.

Qualtrics recorded IP addresses. The original project used Know Your IP to obtain location and blacklist information from external services [Laohaprapanon and Sood, 2018]. This edition uses the deposited metadata. IP addresses and blacklist status change, so querying a current service would measure something different from the original survey conditions. Public record identifiers permit counting repeated IPs without publishing addresses.

The IP screen marks records with missing addresses, a positive blacklist indicator, a repeated recorded address, or a location outside the United States. Repeated-address percentages count *responses* attached to an address that appears more than once. The supplementary multiplicity table instead counts distinct observed addresses and reports how many responses those addresses contribute. Missing addresses are not one shared network. Missing blacklist information is uncertainty about that screen, not affirmative evidence that an address was absent from every blacklist.

The rare-trait screen counts affirmative answers about an artificial limb or prosthetic, vision impairment, hearing impairment, gang membership, immediate-family gang membership, and an unusual amount of sleep. The original threshold was two affirmative answers. The sleep item asks average hours per night and flags the categories "4 or less" and "More than 10". The adjacent "4–6" category overlaps at the boundary, so the recorded category is preserved. The items measure reported experiences, and the combined indicator is a response screen.

The 2020 surveys also asked participants to enter the date. Index 1 combines the IP and rare-trait screens. Index 2 additionally flags the day/month/year convention, and index 3 additionally

flags entries that are not dates. These indices apply increasingly inclusive screens. The date convention measures response format rather than verified location.

The original project reported that respondents passed its reCAPTCHA check. Passing an automated challenge does not establish attentive reading of the questionnaire or exclude every form of automation. Where pre-entry failures were not deposited, the replication cannot estimate the check's failure rate among all attempted entrants.

4 Response patterns across the three surveys

Table 1: Responses flagged by IP metadata

Study	N	Missing	Blacklisted	Repeated	Foreign	Any
August 2018	2000	0.5	16.1	5.3	6.0	20.3
June 2020	1505	0.0	6.7	5.4	1.1	12.1
July 2020	409	0.0	5.9	3.2	0.7	9.8

Percentages of survey records. Categories overlap. A duplicate is a record with an observed IP identifier occurring more than once in its survey. Study 1 rows are paid assignments. These metadata identify screening conditions, not verified deception.

The IP and rare-trait screens capture different aspects of the data. The combined first index flags 24.6% in August 2018, 37.9% in June 2020, and 27.6% in July 2020. The rare-trait screen alone flags 6.2%, 30.0%, and 21.0%, respectively. Table 1 compares the IP components, while Figure 1 separates the rare-trait items. A reduction in IP flags coexists with an increase in rare-trait endorsements. The combined indices in Table 2 report their overlap rather than adding percentages that may count the same record twice.

Table 2: Combined screens and self-reported insincerity

Measure	August 2018	June 2020	July 2020
Rare-trait flagged	6.2	30.0	21.0
Insincerity	8.8	14.4	14.3
Insincerity: some or more	8.8	33.3	28.5
Index 1	24.6	37.9	27.6
Index 2	–	44.6	35.2
Index 3	–	46.2	35.2
Day-first date	–	20.0	16.4
Non-date	–	4.9	4.2

Percentages use observed values of the named measure; unavailable items are shown as –. Insincerity counts some/most/always in 2018 and most/always in 2020; the next row uses some/most/always throughout. Indices 2 and 3 add date-format and non-date flags to index 1. The indices are alternative definitions, not identified bounds on response quality.

Self-reported humorous or insincere responding provides another measure of response quality. Table 2 reports both the study-specific threshold and a common threshold of some of the time or more. Supplementary tables show the association between insincerity and the rare-trait screen, with denominators and tests of association.

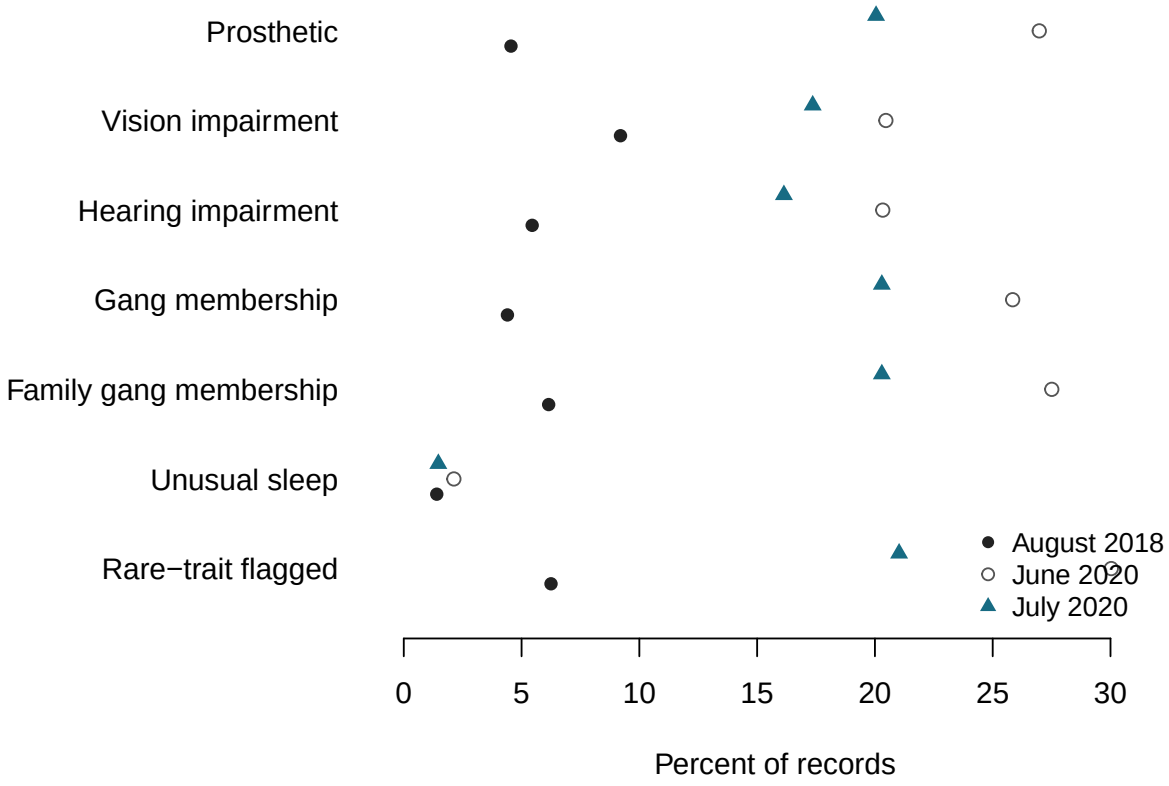


Figure 1: Reported uncommon traits and behaviors across surveys. Points show affirmative percentages under the archived binary recodes, which treat missing rare-trait responses as negative. The final screen combines the archived rare-trait items. These are descriptive sample proportions, not population prevalence estimates.

4.1 Experience and completion times

Respondents in the 2020 surveys reported a category for the number of tasks they had completed. Table 3 reports screen prevalence, self-reported insincerity, and average duration within those categories. The pattern can inform recruitment choices, but it cannot establish what would happen if an otherwise identical requester imposed a new threshold. Experienced respondents may read more efficiently, differ in other characteristics, or recognize familiar designs. Those explanations are not separated by these data.

Table 3: Screening and duration by reported task experience

June 2020				
Measure	Under 100	100–500	500–1,000	Over 1,000
Sample share	21.8	29.0	12.8	36.5
Index 1	56.0	51.5	45.3	14.1
Index 2	68.8	60.2	51.6	15.3
Index 3	70.0	60.9	53.6	17.5
Insincerity	14.7	23.0	16.7	6.6
Duration (seconds)	678.3	608.5	586.2	495.9
July 2020				
Measure	Under 100	100–500	500–1,000	Over 1,000
Sample share	20.7	26.8	9.4	43.1
Index 1	50.0	40.4	31.6	6.9
Index 2	63.1	51.4	34.2	10.9
Index 3	63.1	51.4	34.2	10.9
Insincerity	14.3	24.8	34.2	3.4
Duration (seconds)	544.9	490.8	452.2	280.5

Entries are percentages except mean duration in seconds. Task counts are self-reported categories. Item denominators are recorded in the accompanying CSV.

Timing is also an imperfect proxy for quality. A short duration can reflect speed or familiarity; a long duration can include a break. Supplementary tables report quartiles, median duration, and the original study's timing screens. Missing durations remain missing when classifying slow and fast respondents. Tests that fail to reject a treatment interaction do not show that the groups have equivalent responsiveness: the interval determines which differences remain compatible with the data.

5 Comparisons with other platforms

Table 4 compares the three surveys with earlier MTurk surveys and with SSI/Dynata and Lucid samples [Ahler and Broockman, 2018, Coppock and McClellan, 2019, Graham, 2019, 2020, Thompson and Busby, 2020]. The comparison surveys enter the public analysis as aggregate counts. Their individual records are held by the source projects. Repeated addresses are counted within each

survey, and percentages use all records as the denominator. The table reports unresolved flags separately.

Table 4: IP-flagged responses across surveys (percent)

Survey	Year	Platform	N	Flagged	Unknown
Busby	2019	Lucid	2531	2.84	0
Graham August	2019	Lucid	8555	4.18	0
Graham June	2019	Lucid	7068	2.02	0
Thompson and Busby	2019	Lucid	1798	1.61	0
Graham	2020	Lucid	4233	1.89	1
Ahler and Broockman	2015	MTurk	2045	12.42	0
Ahler and Goggin	2015	MTurk	898	5.68	0
August survey	2018	MTurk	2000	20.30	0
July survey	2020	MTurk	409	9.78	0
June survey	2020	MTurk	1505	12.09	0
Berkeley IGS Poll	2015	SSI/Dynata	2295	4.14	46

Comparison-survey counts are supplied as aggregate inputs. Percentages use all records; Unknown counts unresolved flags. Repeated IPs are counted within each survey.

IP-flag rates are lower in the Lucid comparison surveys than in the three study surveys. Recruitment, questionnaires, and field dates differ across these samples, so the comparison is descriptive. The IP indicators also leave other aspects of response quality, such as attentive reading, unmeasured.

The paper also compared cost per response-minute after removing IP-flagged records. That quantity is payment plus the assumed platform charge, divided by the fraction retained and the assumed survey length. It is a cost per *unflagged* response-minute, not a validated cost per truthful response. Advertised and observed durations answer different questions, and nominal charges in the original study are historical inputs. Modern prices must not be inferred from this calculation.

Table 5: Historical cost per IP-unflagged response-minute

Study	Dollars	Multiplier	Minutes	Flagged	Cents
June 2020	0.75	1.3	15	12.1	7.39
July 2020	0.35	1.3	5	9.8	10.09
Lucid illustrative	1.00	1.0	10	2.0	10.20
Lucid targeted illustrative	2.00	1.0	10	1.6	20.33

Dollars is payment per response. Cents is the cost per retained response-minute, including the historical fee multiplier. Lucid rows preserve illustrative published price, duration, and rounded IP-rate assumptions; they are not updated vendor quotations.

6 Experimental treatment effects and screening

The June 2020 experiment asked partisans to evaluate small decreases in unemployment and inflation. A randomly assigned cue attributed the period either to President Obama or to Republican control of Congress. The analysis codes the cue relative to the respondent's party, including partisan leaners, and compares an out-party cue with an in-party cue. Responses run from worse to better on a scale from zero to one. The questionnaire wording appears in the appendix.

For each outcome, the subgroup model is

$$Y_i = \beta_0 + \beta_1 T_i + \beta_2 F_i + \beta_3 T_i F_i + \varepsilon_i, \tag{1}$$

where T_i indicates the out-party cue and F_i is the specified flag. Figure 2 reports subgroup contrasts on the observed outcome scale, including their uncertainty. Supplementary tables give the corresponding model results. A negative effect means less favorable assessment under the out-party cue.

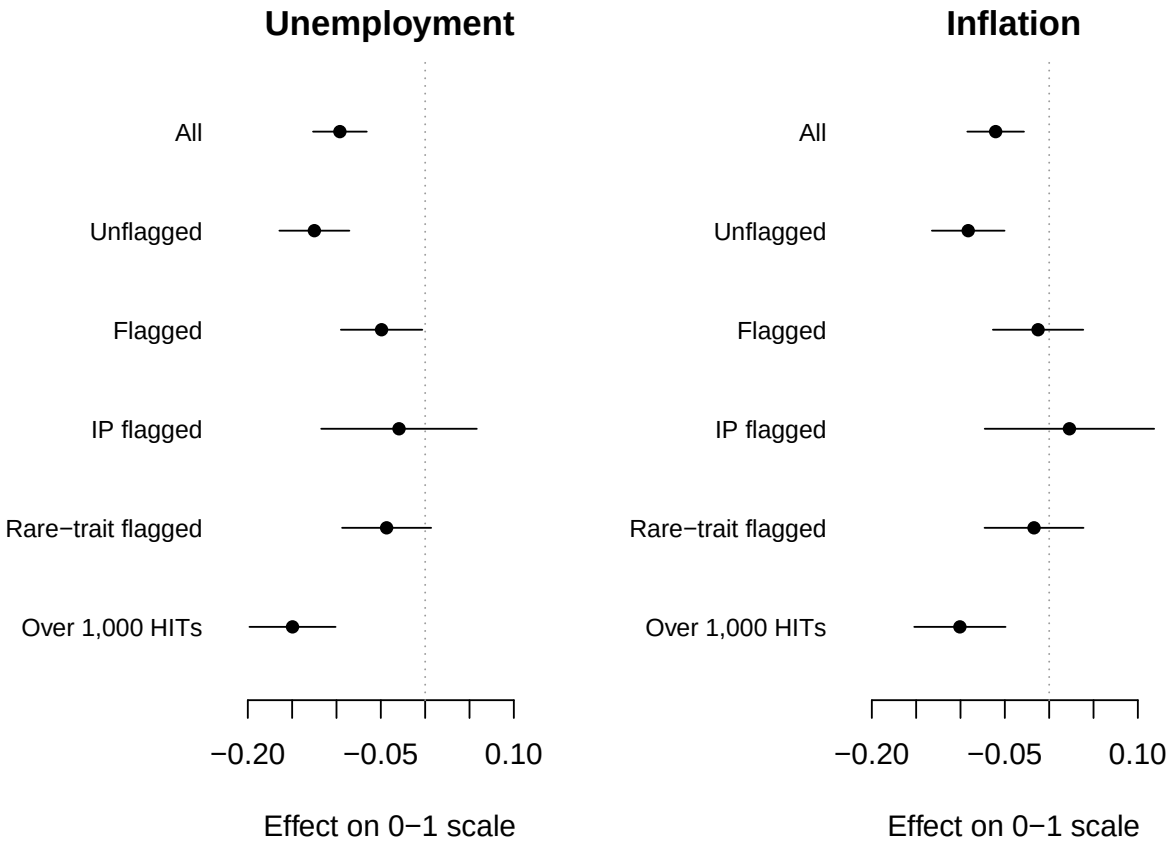


Figure 2: Out-party versus in-party cues in the June 2020 experiment. Points are estimated differences on the zero-to-one economic evaluation scale; lines show 95% confidence intervals from the fitted models. Samples are item-complete partisan respondents within each reported subgroup. Different subgroup point estimates are not themselves a test of the between-group difference.

The full-sample contrasts are -0.096 for unemployment and -0.060 for inflation. The corresponding unflagged contrasts are -0.125 and -0.091; the flagged contrasts are -0.049 and -0.013.

The appendix shows the arm-specific outcome means and endpoint shares, so the treatment differences can be assessed against their baseline response levels. Conventional, heteroskedasticity-robust, and IP-clustered estimates are retained in the generated results; the appendix gives the clustered intervals.

Randomization identifies the effect of the assigned cue within each analysis sample. Screening changes the composition of that sample. The subgroup contrasts show how sensitive the estimates are to the response screens; they do not separately identify an effect of dishonesty.

6.1 What attenuation means here

Additive mean-zero measurement error in an outcome increases uncertainty without necessarily attenuating a randomized treatment contrast. Attenuation instead follows from a particular response-replacement model. Let Y_i^* be a meaningful response, let R_i identify a replacement response, and write

$$Y_i = (1 - R_i)Y_i^* + R_i Z_i. \tag{2}$$

If replacement occurs with probability q independently of treatment and the relevant potential outcomes, and Z_i has the same expectation in each treatment arm, then

$$E[Y_i \mid T_i = 1] - E[Y_i \mid T_i = 0] = (1 - q)\tau, \tag{3}$$

where τ is the meaningful-response treatment contrast. Systematic or treatment-dependent replacement need not move the estimate toward zero. The screen comparisons do not identify q or establish this mechanism.

For the empirical comparison, define the retention ratio as the full-sample contrast divided by the unflagged-sample contrast. Subtracting that ratio from one gives a relative reduction, expressed as a percentage after multiplication by one hundred. Subtracting the contrasts themselves gives a difference on the outcome scale; multiplying that difference by one hundred yields percentage points only when the outcome is a probability or proportion. Ratios are unstable when their denominator is near zero. Because full and unflagged estimates share observations, their uncertainties are dependent. A pooled ratio must state its weights and propagate that dependence.

Table 6: Full-sample and unflagged contrasts: descriptive retention ratios

Study	Contrast	Full	Unflagged	Ratio	Reduction	Lower	Upper
August 2018	Black x -1	0.117	0.140	0.839	16.1	-0.1	31.1
August 2018	Black x 1	0.082	0.097	0.847	15.3	-0.6	30.6
August 2018	Same-sex spouse x -1	0.165	0.191	0.868	13.2	1.5	24.3
August 2018	Same-sex spouse x 1	0.115	0.132	0.876	12.4	0.4	23.6
August 2018	Evangelical x -1	0.044	0.059	0.750	25.0	-49.7	101.3

Study	Contrast	Full	Unflagged	Ratio	Reduction	Lower	Upper
August 2018	Evangelical x 1	0.031	0.042	0.752	24.8	-50.3	101.3
August 2018	Secular x -1	0.067	0.066	1.026	-2.6	-137.5	48.2
August 2018	Secular x 1	0.046	0.045	1.031	-3.1	-134.8	47.8
August 2018	Liberal activity x -1	0.080	0.094	0.852	14.8	-28.8	42.6
August 2018	Liberal activity x 1	0.055	0.064	0.859	14.1	-27.8	41.9
August 2018	Conservative activity x -1	0.088	0.083	1.059	-5.9	-89.5	23.1
August 2018	Conservative activity x 1	0.063	0.059	1.069	-6.9	-92.0	22.9
June 2020	Unemployment	-0.096	-0.125	0.770	23.0	4.2	38.7
June 2020	Inflation	-0.060	-0.091	0.661	33.9	9.2	56.9

Reduction is 100 times one minus the retention ratio. It is a descriptive relative change, not the causal effect of dishonesty. James contrasts are sign-aligned to the unflagged estimate; those signs cancel in the ratios. Lower and Upper are 95 percent percentile intervals for the relative reduction from a paired IP-cluster bootstrap; full and unflagged estimates are refitted in each draw. Respondent-bootstrap results are also supplied in CSV.

7 Implications for survey research

Researchers should examine several response-quality indicators and report how prespecified screens change their estimates. Eligibility, attentive reading, and response consistency require different checks; using several indicators makes the basis for exclusions explicit.

The original recommendations included checking location and repeated submissions, including response-quality items, allowing time to inspect data, considering task experience, and comparing costs across recruitment sources. Those remain design choices to evaluate against their measurement error and effects on sample composition. Participants should know the task and compensation conditions before consenting. Quality bonuses and alternative rating systems are hypotheses about incentives, not interventions tested in these surveys.

IP collection also depends on consent, institutional requirements, and applicable law. Where IP collection is unavailable, other screens can provide information, but neither date conventions nor rare-trait questions are validated substitutes for a participant's location. Any exclusion strategy should report missingness, the records removed, and sensitivity of the substantive findings. The scope of this replication is the archived evidence; the current quality and prices of online recruitment platforms require new data.

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A Supporting analyses

The R pipeline generates all numerical exhibits from the bundled study records and comparison-survey counts. Model intervals concern individual contrasts and do not adjust for simultaneous examination of many outcomes.

The following tables cover the supporting analyses: distributions of observed IP multiplicity and recorded geography; associations between screens; timing distributions and models; subgroup experiments; and ordered-logit models for the James experiment. Country and city labels describe the deposited geolocation service's output. They do not establish residence. The original manual interpretation of the timing of repeated submissions is not reproduced merely by counting duplicates: separating coordinated individuals from one person using several devices requires information these records do not identify.

Table 7: Observed IP multiplicity: addresses and response records

study	multiplicity	n addresses	n responses
August 2018	1	1885	1885
August 2018	2	20	40
August 2018	3	13	39
August 2018	4	4	16
August 2018	5	1	5
August 2018	6	1	6
June 2020	1	1424	1424
June 2020	2	33	66
June 2020	3	3	9
June 2020	6	1	6
July 2020	1	396	396
July 2020	2	5	10
July 2020	3	1	3

Table 8: Response records by recorded country

study	location	n
August 2018	Canada	6
August 2018	India	17
August 2018	Other	54
August 2018	United States	1870
August 2018	Venezuela	42
June 2020	India	12
June 2020	Other	5
June 2020	United States	1488
July 2020	Other	3
July 2020	United States	406

Table 9: Response records by recorded city

study	location	n
August 2018	Buffalo	77
August 2018	Kansas City	28
August 2018	Los Angeles	44
August 2018	Maracaibo	31
August 2018	New York	72
August 2018	Other	1601
June 2020	Chicago	40
June 2020	Las Vegas	35
June 2020	Los Angeles	52
June 2020	New York	31
June 2020	Other	1224
July 2020	Other	384

Table 10: Sensitivity of screen counts to missing rare-trait items

Study	N	Missing	RareMin	RareMax	IndexMin	IndexMax
August 2018	2000	9	125	134	493	493
June 2020	1505	4	452	455	571	574
July 2020	409	2	86	88	113	113

Minimum sets missing rare-trait items to negative; maximum sets them to affirmative. These bound the screen under missing answers, not the number of dishonest participants.

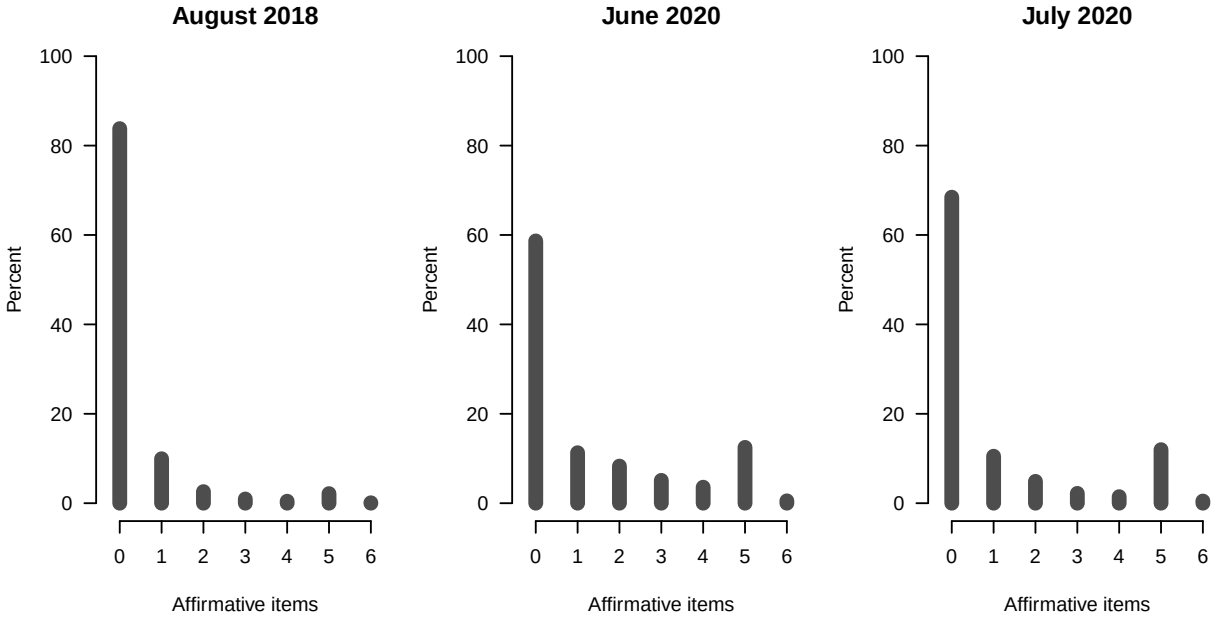


Figure 3: Distribution of the number of affirmative rare-trait items. Each panel uses the archived item recodes and the same percentage scale.

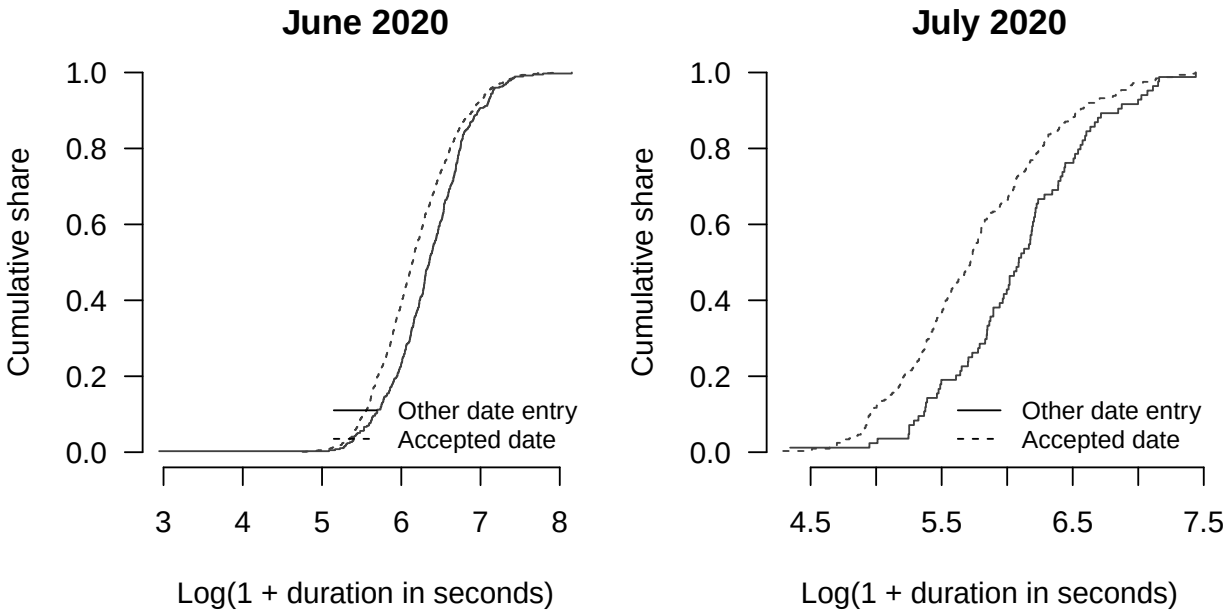


Figure 4: Duration distributions by date convention in the two 2020 surveys. Empirical cumulative distributions compare accepted date entries with other entries, including non-dates, and retain long completion times on a logarithmic time axis. Date convention does not establish location or citizenship.

Table 11: Associations with the rare-trait screen

Study	Measure	Screen	N	Percent	ChiSq	p
August 2018	Insincerity	Not rare-trait flagged	1865	6.6	179.0	<0.001
August 2018	Insincerity	Rare-trait flagged	125	41.6	179.0	<0.001
August 2018	IP flagged	Not rare-trait flagged	1875	19.6	8.4	0.004
August 2018	IP flagged	Rare-trait flagged	125	30.4	8.4	0.004
June 2020	Insincerity	Not rare-trait flagged	1051	10.2	49.9	<0.001
June 2020	Insincerity	Rare-trait flagged	452	24.1	49.9	<0.001
June 2020	IP flagged	Not rare-trait flagged	1053	11.3	2.1	0.150
June 2020	IP flagged	Rare-trait flagged	452	13.9	2.1	0.150
July 2020	Insincerity	Not rare-trait flagged	321	12.1	5.5	0.019
July 2020	Insincerity	Rare-trait flagged	86	22.1	5.5	0.019
July 2020	IP flagged	Not rare-trait flagged	323	8.4	3.5	0.061
July 2020	IP flagged	Rare-trait flagged	86	15.1	3.5	0.061

Table 12: Survey duration in seconds by screening group

Study	Group	N	Mean	SD	Q1	Median	Q3
August 2018	All	1991	679.5	410.4	426.0	573.0	785.0
August 2018	Flagged	484	805.1	497.7	467.5	659.5	1017.0
August 2018	Unflagged	1507	639.2	369.4	420.0	554.0	727.0
June 2020	All	1505	579.1	345.4	349.0	488.0	721.0

Study	Group	N	Mean	SD	Q1	Median	Q3
June 2020	Flagged	571	615.9	339.4	393.0	543.0	767.0
June 2020	Unflagged	934	556.6	347.2	321.0	455.0	670.0
June 2020	Day-first date	301	651.9	325.8	421.0	579.0	802.0
June 2020	Index 2 flagged	671	636.9	352.5	401.0	555.0	787.0
June 2020	Index 3 flagged	696	631.0	351.0	399.5	548.5	782.0
July 2020	All	409	405.7	280.8	215.0	322.0	500.0
July 2020	Flagged	113	513.1	315.3	318.0	429.0	641.0
July 2020	Unflagged	296	364.7	255.3	200.0	290.0	440.0
July 2020	Day-first date	67	543.9	318.7	322.0	475.0	692.0
July 2020	Index 2 flagged	144	526.6	316.2	320.0	442.5	665.5
July 2020	Index 3 flagged	144	526.6	316.2	320.0	442.5	665.5

Table 13: Timing regressions: differences associated with each screen

Study	Outcome	Screen	Estimate	SE	N
August 2018	Duration (seconds)	Index 1	165.864	21.122	1991
August 2018	Fast timing screen	Index 1	-0.003	0.007	1991
August 2018	Slow timing screen	Index 1	0.124	0.016	1991
June 2020	Duration (seconds)	Index 1	59.213	18.289	1505
June 2020	Fast timing screen	Index 1	-0.024	0.013	1505
June 2020	Slow timing screen	Index 1	-0.012	0.014	1505
June 2020	Duration (seconds)	Index 2	104.370	17.713	1505
June 2020	Fast timing screen	Index 2	-0.034	0.013	1505
June 2020	Slow timing screen	Index 2	0.016	0.014	1505
June 2020	Duration (seconds)	Index 3	96.605	17.687	1505
June 2020	Fast timing screen	Index 3	-0.022	0.013	1505
June 2020	Slow timing screen	Index 3	0.013	0.014	1505
June 2020	Duration (seconds)	Day-first date	90.938	22.140	1505
June 2020	Fast timing screen	Day-first date	-0.034	0.016	1505
June 2020	Slow timing screen	Day-first date	0.022	0.017	1505
July 2020	Duration (seconds)	Index 1	148.381	30.206	409
July 2020	Fast timing screen	Index 1	-0.129	0.039	409
July 2020	Slow timing screen	Index 1	0.041	0.031	409
July 2020	Duration (seconds)	Index 2	186.576	27.597	409
July 2020	Fast timing screen	Index 2	-0.162	0.036	409
July 2020	Slow timing screen	Index 2	0.082	0.029	409
July 2020	Duration (seconds)	Index 3	186.576	27.597	409
July 2020	Fast timing screen	Index 3	-0.162	0.036	409
July 2020	Slow timing screen	Index 3	0.082	0.029	409
July 2020	Duration (seconds)	Day-first date	165.187	36.657	409
July 2020	Fast timing screen	Day-first date	-0.158	0.047	409
July 2020	Slow timing screen	Day-first date	0.094	0.037	409
July 2020	Duration (seconds)	Date convention	165.032	37.289	392
July 2020	Fast timing screen	Date convention	-0.160	0.047	392
July 2020	Slow timing screen	Date convention	0.090	0.038	392

Table 14: Date-convention groups: duration distribution comparison

study	n accepted	n other	statistic	p value
June 2020	1130	375	0.1813	<0.001
July 2020	325	84	0.3266	<0.001

Table 15: Economic experiment: subgroup contrasts (conventional SEs)

Outcome	Group	Term	Estimate	SE	N
Unemployment	All	Out-party cue	-0.096	0.015	1425
Inflation	All	Out-party cue	-0.060	0.016	1425
Unemployment	Unflagged	Out-party cue	-0.125	0.020	861
Inflation	Unflagged	Out-party cue	-0.091	0.021	861
Unemployment	Flagged	Out-party cue	-0.049	0.023	564
Inflation	Flagged	Out-party cue	-0.013	0.026	564
Unemployment	IP flagged	Out-party cue	-0.029	0.044	182
Inflation	IP flagged	Out-party cue	0.023	0.048	182
Unemployment	Rare-trait flagged	Out-party cue	-0.043	0.026	445
Inflation	Rare-trait flagged	Out-party cue	-0.017	0.028	445
Unemployment	Over 1,000 HITs	Out-party cue	-0.150	0.025	505
Inflation	Over 1,000 HITs	Out-party cue	-0.101	0.026	505

Table 16: Economic experiment: screening and timing interactions

Outcome	Group	Term	Estimate	SE	N
Unemployment	Index 1	Out-party cue	-0.125	0.020	1425
Unemployment	Index 1	Index 1	0.045	0.022	1425
Unemployment	Index 1	Out-party cue x Index 1	0.076	0.031	1425
Inflation	Index 1	Out-party cue	-0.091	0.021	1425
Inflation	Index 1	Index 1	-0.022	0.024	1425
Inflation	Index 1	Out-party cue x Index 1	0.079	0.033	1425
Unemployment	Fast timing screen	Out-party cue	-0.097	0.016	1425
Unemployment	Fast timing screen	Fast timing screen	0.012	0.047	1425
Unemployment	Fast timing screen	Out-party cue x Fast timing screen	0.011	0.064	1425
Inflation	Fast timing screen	Out-party cue	-0.063	0.017	1425
Inflation	Fast timing screen	Fast timing screen	-0.004	0.049	1425
Inflation	Fast timing screen	Out-party cue x Fast timing screen	0.043	0.067	1425

Table 17: James experiment: linear probability specifications

Outcome	Group	Term	Estimate	SE	N
Democratic conjunction	All	Black	0.101	0.022	1991
Democratic conjunction	All	Same-sex spouse	0.150	0.022	1991
Democratic conjunction	All	Evangelical	-0.042	0.026	1991
Democratic conjunction	All	Secular	0.063	0.027	1991
Democratic conjunction	All	Liberal activity	0.089	0.026	1991
Democratic conjunction	All	Conservative activity	-0.077	0.027	1991
Republican conjunction	All	Black	-0.109	0.018	1991
Republican conjunction	All	Same-sex spouse	-0.139	0.018	1991
Republican conjunction	All	Evangelical	0.037	0.022	1991
Republican conjunction	All	Secular	-0.048	0.022	1991
Republican conjunction	All	Liberal activity	-0.053	0.022	1991
Republican conjunction	All	Conservative activity	0.074	0.022	1991

Table 18: Economic outcome levels by treatment arm: means and endpoint percentages

Group	Outcome	Cue	N	Mean	Zero	One
All	Unemployment	In-party	689	0.793	3.3	62.0
All	Unemployment	Out-party	736	0.697	6.4	45.8
All	Inflation	In-party	689	0.711	6.1	48.3
All	Inflation	Out-party	736	0.651	8.8	39.0
Unflagged	Unemployment	In-party	411	0.775	4.1	59.1
Unflagged	Unemployment	Out-party	450	0.650	7.6	37.6
Unflagged	Inflation	In-party	411	0.720	6.6	50.6
Unflagged	Inflation	Out-party	450	0.629	8.9	34.7
Flagged	Unemployment	In-party	278	0.820	2.2	66.2
Flagged	Unemployment	Out-party	286	0.771	4.5	58.7
Flagged	Inflation	In-party	278	0.698	5.4	45.0
Flagged	Inflation	Out-party	286	0.685	8.7	45.8

Table 19: Economic experiment: 95 percent intervals with IP-clustered inference

Group	Outcome	N	Clusters	Estimate	Lower	Upper
All	Unemployment	1425	1381	-0.096	-0.127	-0.066
All	Inflation	1425	1381	-0.060	-0.092	-0.029
Unflagged	Unemployment	861	861	-0.125	-0.164	-0.086
Unflagged	Inflation	861	861	-0.091	-0.132	-0.050
Flagged	Unemployment	564	520	-0.049	-0.096	-0.002
Flagged	Inflation	564	520	-0.013	-0.063	0.038

B The James experiment

The August 2018 experiment varied a fictional person's race, spouse's name, political activity, and religious activity. Participants chose whether James was most likely a salesman, a salesman who supported Democrats, or a salesman who supported Republicans. Choosing either conjunction over the unrestricted salesman description violates the probability ordering. It does not make the conjunction event itself logically impossible. The experiment examines partisan stereotyping through those judgments [Ahler and Sood, 2017, 2018, Tversky and Kahneman, 1974].

The original analysis orders the three outcomes as Democratic conjunction, unrestricted occupation, and Republican conjunction, then fits a proportional-odds model with randomized attributes and their interactions with the relevant screen. That ordering is a modeling assumption about political direction. It is not an ordering from least to most accurate response. For cutpoint c_j and linear predictor η_i, the model uses

$$\Pr(Y_i \leq j) = \text{logit}^{-1}(c_j - \eta_i). \tag{4}$$

Reported effects change an attribute from its reference category, calculate the difference in each outcome probability, and average over the specified empirical covariate distribution. They are finite changes for categorical treatments, not derivatives of a continuous exposure. The appendix reports coefficients and probability changes with their uncertainty, including the timing comparison. The reported margins change one indicator while averaging over the observed values of the other attributes.

Table 20: James experiment: ordered-logit coefficients and cutpoints

Group	Term	Estimate	SE	N
All	Black	-0.500	0.087	1991
All	Same-sex spouse	-0.698	0.088	1991
All	Evangelical	0.188	0.105	1991
All	Secular	-0.288	0.109	1991
All	Liberal activity	-0.343	0.107	1991
All	Conservative activity	0.373	0.105	1991
All	-1|0	-0.553	0.115	1991
All	0|1	0.704	0.116	1991

Group	Term	Estimate	SE	N
Unflagged	Black	-0.609	0.102	1507
Unflagged	Same-sex spouse	-0.821	0.103	1507
Unflagged	Evangelical	0.256	0.122	1507
Unflagged	Secular	-0.288	0.127	1507
Unflagged	Liberal activity	-0.413	0.126	1507
Unflagged	Conservative activity	0.360	0.123	1507
Unflagged	-1|0	-0.578	0.134	1507
Unflagged	0|1	0.644	0.135	1507
Flagged	Black	-0.210	0.172	484
Flagged	Same-sex spouse	-0.383	0.172	484
Flagged	Evangelical	-0.053	0.208	484
Flagged	Secular	-0.303	0.212	484
Flagged	Liberal activity	-0.107	0.210	484
Flagged	Conservative activity	0.484	0.207	484
Flagged	-1|0	-0.500	0.226	484
Flagged	0|1	0.891	0.229	484
IP flagged	Black	-0.312	0.193	397
IP flagged	Same-sex spouse	-0.482	0.191	397
IP flagged	Evangelical	0.081	0.231	397
IP flagged	Secular	-0.235	0.235	397
IP flagged	Liberal activity	-0.229	0.233	397
IP flagged	Conservative activity	0.518	0.232	397
IP flagged	-1|0	-0.518	0.246	397
IP flagged	0|1	0.880	0.250	397
Rare-trait flagged	Black	0.202	0.341	125
Rare-trait flagged	Same-sex spouse	0.025	0.340	125
Rare-trait flagged	Evangelical	-0.215	0.421	125
Rare-trait flagged	Secular	-0.527	0.417	125
Rare-trait flagged	Liberal activity	-0.044	0.426	125
Rare-trait flagged	Conservative activity	0.724	0.409	125
Rare-trait flagged	-1|0	-0.340	0.447	125
Rare-trait flagged	0|1	1.283	0.463	125
All-flag interaction	Black	-0.616	0.103	1991
All-flag interaction	Same-sex spouse	-0.830	0.103	1991
All-flag interaction	Evangelical	0.259	0.122	1991
All-flag interaction	Secular	-0.291	0.128	1991
All-flag interaction	Liberal activity	-0.417	0.126	1991
All-flag interaction	Conservative activity	0.364	0.123	1991
All-flag interaction	Flagged	-0.154	0.256	1991
All-flag interaction	Black x Flagged	0.413	0.199	1991
All-flag interaction	Same-sex spouse x Flagged	0.461	0.198	1991
All-flag interaction	Evangelical x Flagged	-0.310	0.240	1991
All-flag interaction	Secular x Flagged	-0.001	0.246	1991
All-flag interaction	Liberal activity x Flagged	0.313	0.243	1991
All-flag interaction	Conservative activity x Flagged	0.102	0.239	1991

Group	Term	Estimate	SE	N
All-flag interaction	-1|0	-0.600	0.134	1991
All-flag interaction	0|1	0.667	0.134	1991
IP-only interaction	Black	-0.612	0.102	1866
IP-only interaction	Same-sex spouse	-0.825	0.103	1866
IP-only interaction	Evangelical	0.257	0.122	1866
IP-only interaction	Secular	-0.289	0.127	1866
IP-only interaction	Liberal activity	-0.415	0.126	1866
IP-only interaction	Conservative activity	0.362	0.123	1866
IP-only interaction	IP flagged	-0.112	0.289	1866
IP-only interaction	Black x IP flagged	0.281	0.226	1866
IP-only interaction	Same-sex spouse x IP flagged	0.337	0.224	1866
IP-only interaction	Evangelical x IP flagged	-0.241	0.271	1866
IP-only interaction	Secular x IP flagged	0.064	0.278	1866
IP-only interaction	Liberal activity x IP flagged	0.310	0.275	1866
IP-only interaction	Conservative activity x IP flagged	0.091	0.273	1866
IP-only interaction	-1|0	-0.589	0.134	1866
IP-only interaction	0|1	0.655	0.134	1866
Rare-only interaction	Black	-0.611	0.102	1594
Rare-only interaction	Same-sex spouse	-0.824	0.103	1594
Rare-only interaction	Evangelical	0.257	0.122	1594
Rare-only interaction	Secular	-0.289	0.127	1594
Rare-only interaction	Liberal activity	-0.414	0.126	1594
Rare-only interaction	Conservative activity	0.361	0.123	1594
Rare-only interaction	Rare-trait flagged	-0.280	0.590	1594
Rare-only interaction	Black x Rare-trait flagged	0.854	0.423	1594
Rare-only interaction	Same-sex spouse x Rare-trait flagged	0.946	0.423	1594
Rare-only interaction	Evangelical x Rare-trait flagged	-0.768	0.540	1594
Rare-only interaction	Secular x Rare-trait flagged	-0.424	0.525	1594
Rare-only interaction	Liberal activity x Rare-trait flagged	0.950	0.514	1594
Rare-only interaction	Conservative activity x Rare-trait flagged	0.211	0.506	1594
Rare-only interaction	-1|0	-0.586	0.134	1594
Rare-only interaction	0|1	0.652	0.135	1594
Fast, unflagged	Black	-0.617	0.103	1507
Fast, unflagged	Same-sex spouse	-0.826	0.104	1507
Fast, unflagged	Evangelical	0.259	0.123	1507
Fast, unflagged	Secular	-0.259	0.129	1507
Fast, unflagged	Liberal activity	-0.411	0.127	1507
Fast, unflagged	Conservative activity	0.372	0.124	1507
Fast, unflagged	Fast timing screen	0.551	1.038	1507
Fast, unflagged	Black x Fast timing screen	0.975	0.970	1507

Group	Term	Estimate	SE	N
Fast, unflagged	Same-sex spouse x Fast timing screen	-0.128	0.859	1507
Fast, unflagged	Evangelical x Fast timing screen	-0.519	1.002	1507
Fast, unflagged	Secular x Fast timing screen	-1.836	1.078	1507
Fast, unflagged	Liberal activity x Fast timing screen	-0.547	1.056	1507
Fast, unflagged	Conservative activity x Fast timing screen	-0.989	0.957	1507
Fast, unflagged	-1|0	-0.573	0.136	1507
Fast, unflagged	0|1	0.652	0.136	1507

Table 21: James experiment: average discrete changes in outcome probabilities (percentage points)

Group	Attribute	Outcome	Change	SE	N
All	Black	Democratic	11.71	2.03	1991
All	Black	Sales	-3.49	0.65	1991
All	Black	Republican	-8.23	1.44	1991
All	Same-sex spouse	Democratic	16.54	2.04	1991
All	Same-sex spouse	Sales	-5.01	0.70	1991
All	Same-sex spouse	Republican	-11.53	1.45	1991
All	Evangelical	Democratic	-4.39	2.45	1991
All	Evangelical	Sales	1.26	0.69	1991
All	Evangelical	Republican	3.13	1.77	1991
All	Secular	Democratic	6.73	2.53	1991
All	Secular	Sales	-2.11	0.85	1991
All	Secular	Republican	-4.61	1.69	1991
All	Liberal activity	Democratic	8.03	2.51	1991
All	Liberal activity	Sales	-2.55	0.87	1991
All	Liberal activity	Republican	-5.48	1.67	1991
All	Conservative activity	Democratic	-8.77	2.48	1991
All	Conservative activity	Sales	2.43	0.66	1991
All	Conservative activity	Republican	6.34	1.85	1991
Unflagged	Black	Democratic	13.96	2.31	1507
Unflagged	Black	Sales	-4.24	0.77	1507
Unflagged	Black	Republican	-9.72	1.63	1507
Unflagged	Same-sex spouse	Democratic	19.07	2.32	1507
Unflagged	Same-sex spouse	Sales	-5.90	0.85	1507
Unflagged	Same-sex spouse	Republican	-13.17	1.65	1507
Unflagged	Evangelical	Democratic	-5.86	2.79	1507
Unflagged	Evangelical	Sales	1.69	0.79	1507
Unflagged	Evangelical	Republican	4.17	2.02	1507
Unflagged	Secular	Democratic	6.56	2.89	1507
Unflagged	Secular	Sales	-2.08	0.98	1507
Unflagged	Secular	Republican	-4.47	1.93	1507

Group	Attribute	Outcome	Change	SE	N
Unflagged	Liberal activity	Democratic	9.43	2.85	1507
Unflagged	Liberal activity	Sales	-3.05	1.01	1507
Unflagged	Liberal activity	Republican	-6.37	1.87	1507
Unflagged	Conservative activity	Democratic	-8.28	2.84	1507
Unflagged	Conservative activity	Sales	2.35	0.78	1507
Unflagged	Conservative activity	Republican	5.93	2.09	1507
Flagged	Black	Democratic	5.06	4.12	484
Flagged	Black	Sales	-1.34	1.10	484
Flagged	Black	Republican	-3.72	3.05	484
Flagged	Same-sex spouse	Democratic	9.26	4.13	484
Flagged	Same-sex spouse	Sales	-2.46	1.17	484
Flagged	Same-sex spouse	Republican	-6.79	3.06	484
Flagged	Evangelical	Democratic	1.28	4.99	484
Flagged	Evangelical	Sales	-0.34	1.36	484
Flagged	Evangelical	Republican	-0.94	3.64	484
Flagged	Secular	Democratic	7.31	5.13	484
Flagged	Secular	Sales	-2.11	1.62	484
Flagged	Secular	Republican	-5.20	3.55	484
Flagged	Liberal activity	Democratic	2.59	5.09	484
Flagged	Liberal activity	Sales	-0.71	1.45	484
Flagged	Liberal activity	Republican	-1.88	3.64	484
Flagged	Conservative activity	Democratic	-11.63	4.90	484
Flagged	Conservative activity	Sales	2.66	1.08	484
Flagged	Conservative activity	Republican	8.97	4.00	484
IP flagged	Black	Democratic	7.45	4.57	397
IP flagged	Black	Sales	-2.11	1.33	397
IP flagged	Black	Republican	-5.34	3.30	397
IP flagged	Same-sex spouse	Democratic	11.59	4.55	397
IP flagged	Same-sex spouse	Sales	-3.32	1.41	397
IP flagged	Same-sex spouse	Republican	-8.27	3.30	397
IP flagged	Evangelical	Democratic	-1.92	5.49	397
IP flagged	Evangelical	Sales	0.54	1.51	397
IP flagged	Evangelical	Republican	1.38	3.98	397
IP flagged	Secular	Democratic	5.63	5.62	397
IP flagged	Secular	Sales	-1.70	1.81	397
IP flagged	Secular	Republican	-3.92	3.83	397
IP flagged	Liberal activity	Democratic	5.51	5.63	397
IP flagged	Liberal activity	Sales	-1.67	1.83	397
IP flagged	Liberal activity	Republican	-3.83	3.83	397
IP flagged	Conservative activity	Democratic	-12.39	5.46	397
IP flagged	Conservative activity	Sales	3.08	1.30	397
IP flagged	Conservative activity	Republican	9.32	4.36	397
Rare-trait flagged	Black	Democratic	-4.64	7.85	125
Rare-trait flagged	Black	Sales	1.08	1.91	125
Rare-trait flagged	Black	Republican	3.56	6.01	125
Rare-trait flagged	Same-sex spouse	Democratic	-0.58	7.79	125

Group	Attribute	Outcome	Change	SE	N
Rare-trait flagged	Same-sex spouse	Sales	0.13	1.77	125
Rare-trait flagged	Same-sex spouse	Republican	0.45	6.03	125
Rare-trait flagged	Evangelical	Democratic	4.94	9.61	125
Rare-trait flagged	Evangelical	Sales	-1.19	2.46	125
Rare-trait flagged	Evangelical	Republican	-3.75	7.20	125
Rare-trait flagged	Secular	Democratic	12.38	9.83	125
Rare-trait flagged	Secular	Sales	-3.41	3.33	125
Rare-trait flagged	Secular	Republican	-8.97	6.79	125
Rare-trait flagged	Liberal activity	Democratic	1.00	9.84	125
Rare-trait flagged	Liberal activity	Sales	-0.23	2.35	125
Rare-trait flagged	Liberal activity	Republican	-0.77	7.49	125
Rare-trait flagged	Conservative activity	Democratic	-16.40	8.81	125
Rare-trait flagged	Conservative activity	Sales	2.68	1.95	125
Rare-trait flagged	Conservative activity	Republican	13.73	8.18	125

C Question wording

The wording and design descriptions below reproduce the original manuscript's instruments. Quotation marks identify verbatim items; unquoted passages summarize instructions. Numbers in quoted stimuli are design inputs, not estimates from the replication.

C.1 Rare-trait battery and sincerity

The yes/no items were: "Do you use an artificial limb or prosthetic?"; "Are you blind or do you have vision impairment?"; "Are you deaf or do you have hearing impairment?"; "Are you in a gang?"; and "Is one or more of your immediate family members in a gang?" The sleep item reads: "On average, how many hours do you sleep each night?" Options were "4 or less," "4–6," "6–8," "8–10," and "More than 10." These choices and their codes agree across the three locally supplied instruments. The item text and source hashes are recorded in the accompanying instrument documentation.

The sincerity item read: "Finally, we sometimes find people don't always take surveys seriously, instead of providing humorous or insincere responses to questions. How often do you do this?" Options were never, rarely, some of the time, most of the time, and always. The study-specific insincerity measure counts some, most, or always in Study 1, but only most or always in Studies 2 and 3. The harmonized comparison counts some, most, or always in every study. Missing responses are excluded from each denominator.

C.2 Task experience and the date

Respondents were asked to visit their worker qualifications page and find their "Total Approved HITs" number, or give a best guess if unable to find it. The question was "About how many HITs have you completed on MTurk?" Options were fewer than 100; between 100 and 500; between 500 and 1,000; and more than 1,000. The original response categories have overlapping boundary language; the replication preserves the recorded category rather than inventing exact counts.

The date prompt was: "Please write today's date in the text box below. Be sure to type it using the following format: 00/00/0000." It did not explicitly request the US convention.

C.3 Economic experiment

The introductory cue referred either to Barack Obama being president or to Republicans controlling both Houses of Congress. The unemployment item said: "During 2016, [cue], unemployment decreased from

5.0% to 4.8%, a change of 0.2 percentage points. How would you interpret this change? Would you say that unemployment got better, stayed about the same, or got worse?" The inflation item said that inflation decreased from 2.1% to 1.9%, also a change of 0.2 percentage points, and asked the same evaluation. Options were got better, stayed the same, and got worse.

C.4 James vignette

The manipulation read as follows, with alternatives separated by slashes:

> James is a 37-year-old (white / black) man. He attended the University of Michigan, where he double-majored in economics and political science. While there, James was president of a business and marketing club. He also participated in (anti-tax demonstrations / living-wage demonstrations / student government).
>
> James's co-workers describe him as highly driven, outspoken, and confident. He is married to (Karen / Keith) and has one son. In James's free time, he (leads his son's Cub Scouts group, organized through the Baptist Church the family attends / leads his son's Junior Explorers group, led through the Secular Families Foundation / coaches his son's youth sports teams).

The grade-point-average question was: "What do you think James' GPA was in college?" The categories were 3.80–4.00; 3.50–3.79; 3.00–3.49; 2.50–2.99; and 2.49 or below. The conjunction question asked: "Which of the following do you think is most likely?" Options were "James works in sales"; "James works in sales and is an active supporter of the Democratic Party"; and "James works in sales and is an active supporter of the Republican Party." The randomized attribute comparisons preserve the reference categories in the deposited code.